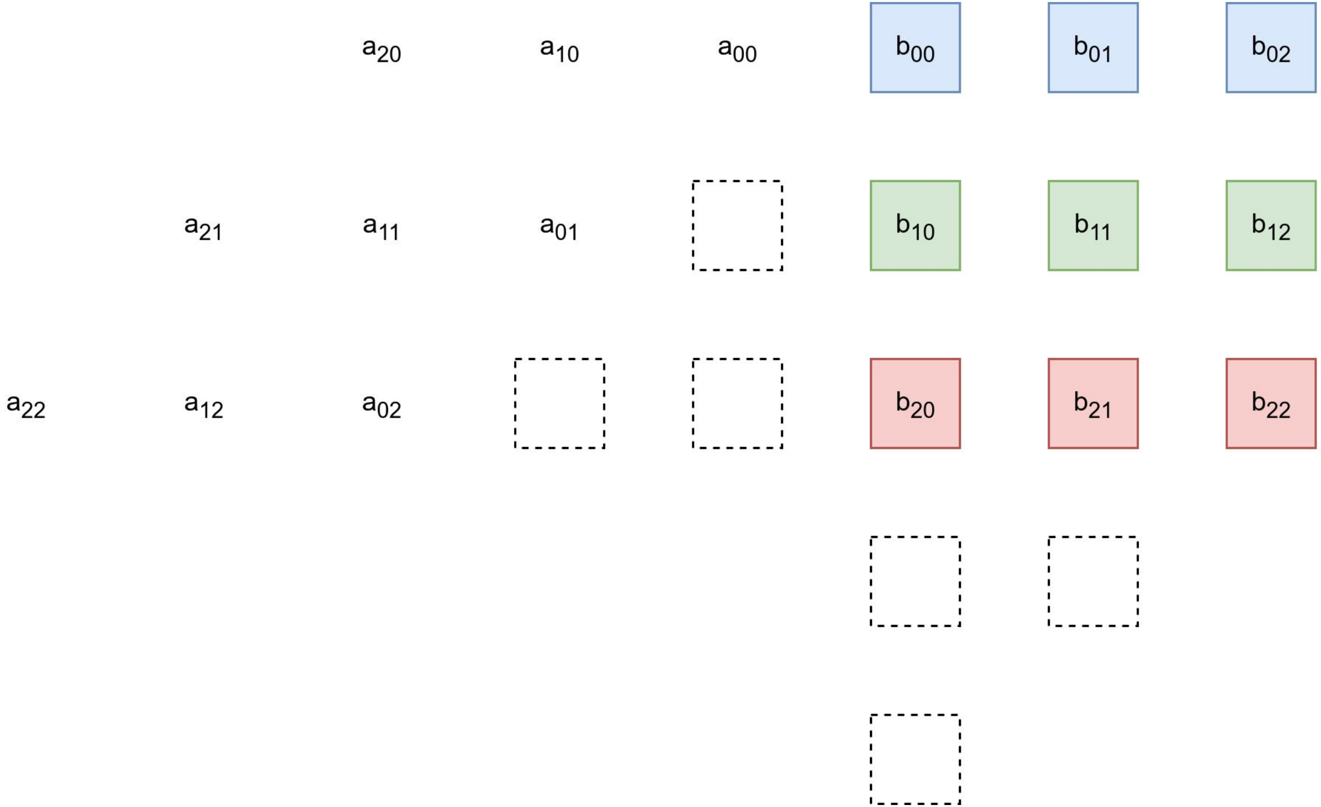
Систолический массив

Проектирование цифровых вычислительных систем

Прутьянов В. В.

Systolic Array



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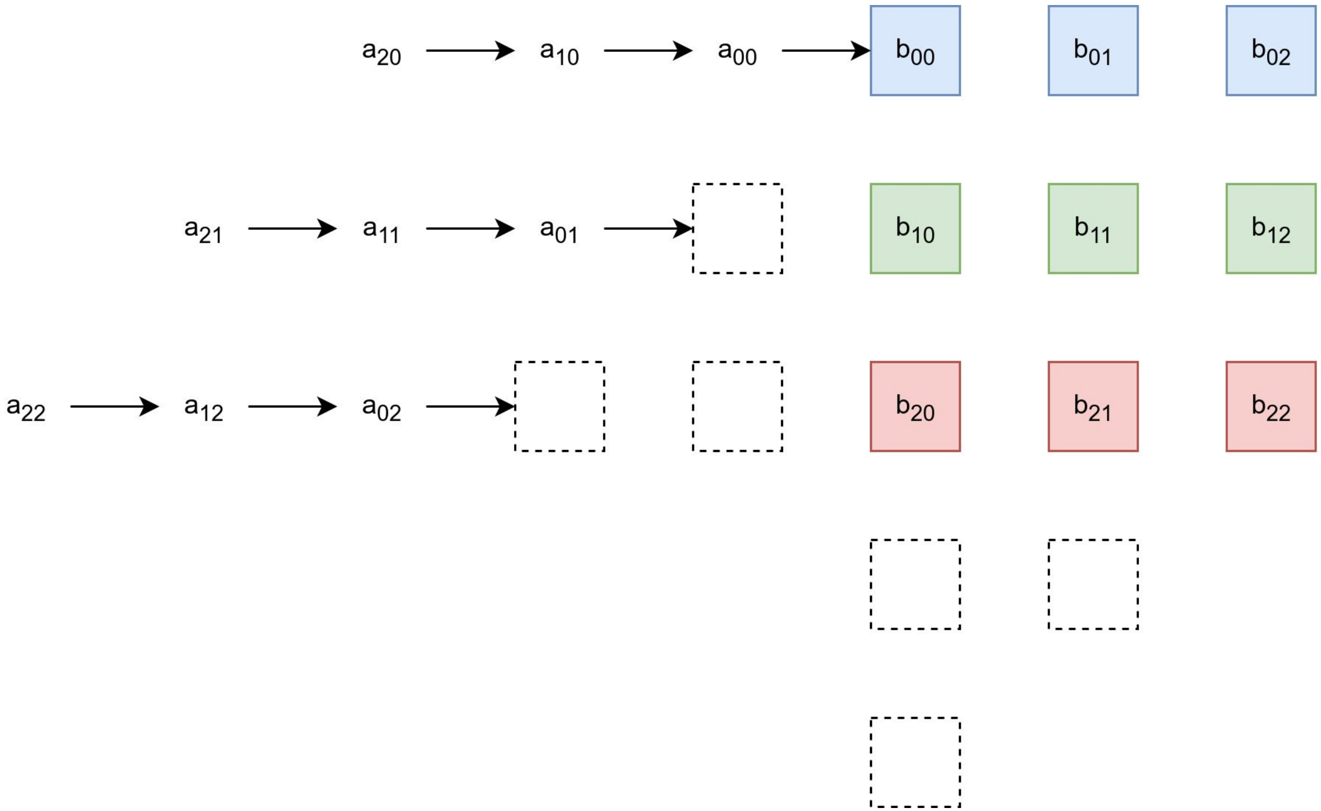

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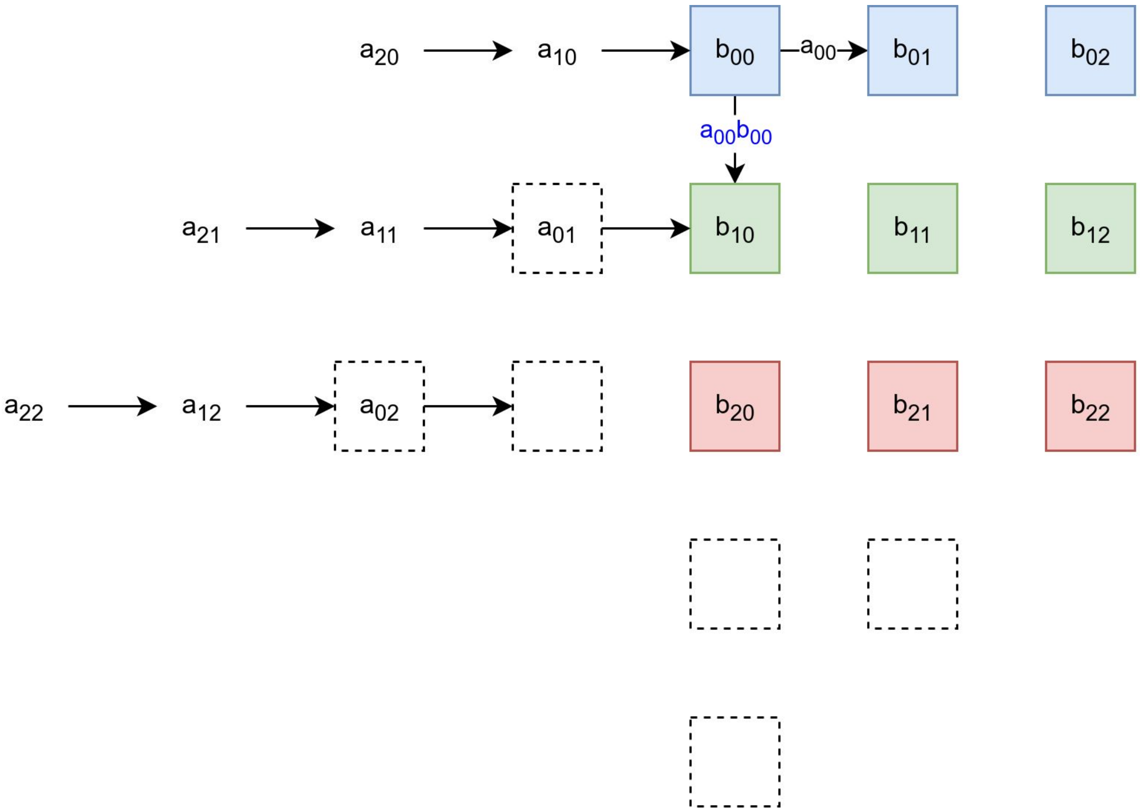

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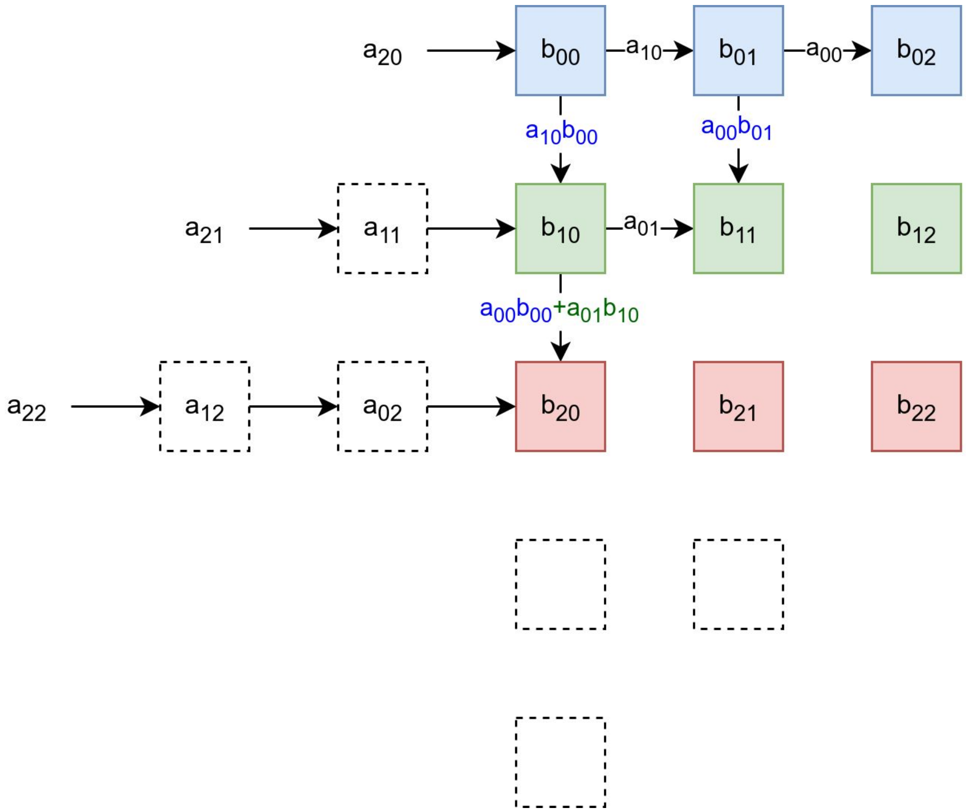

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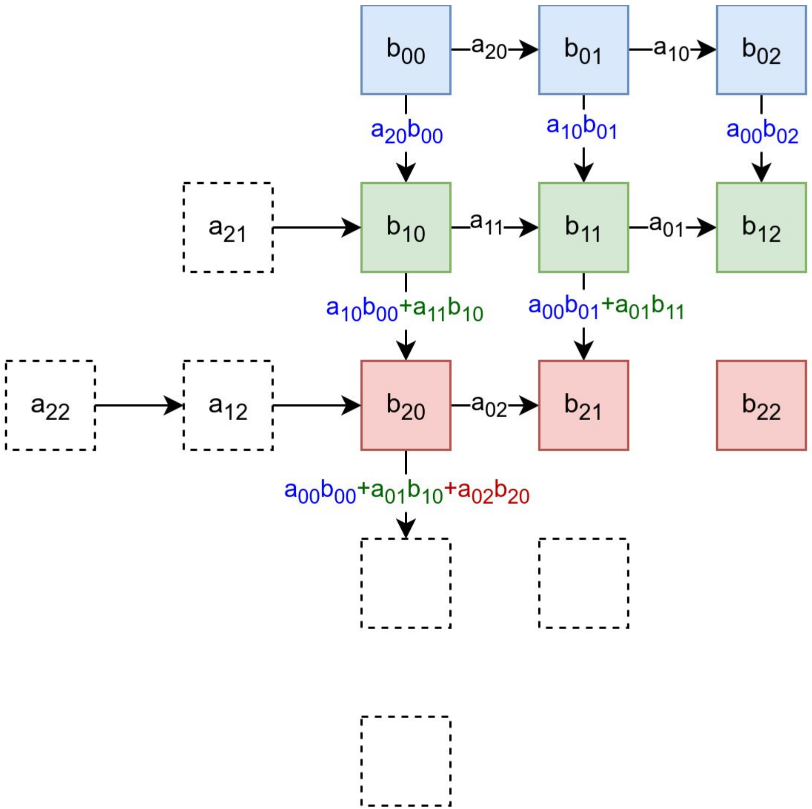

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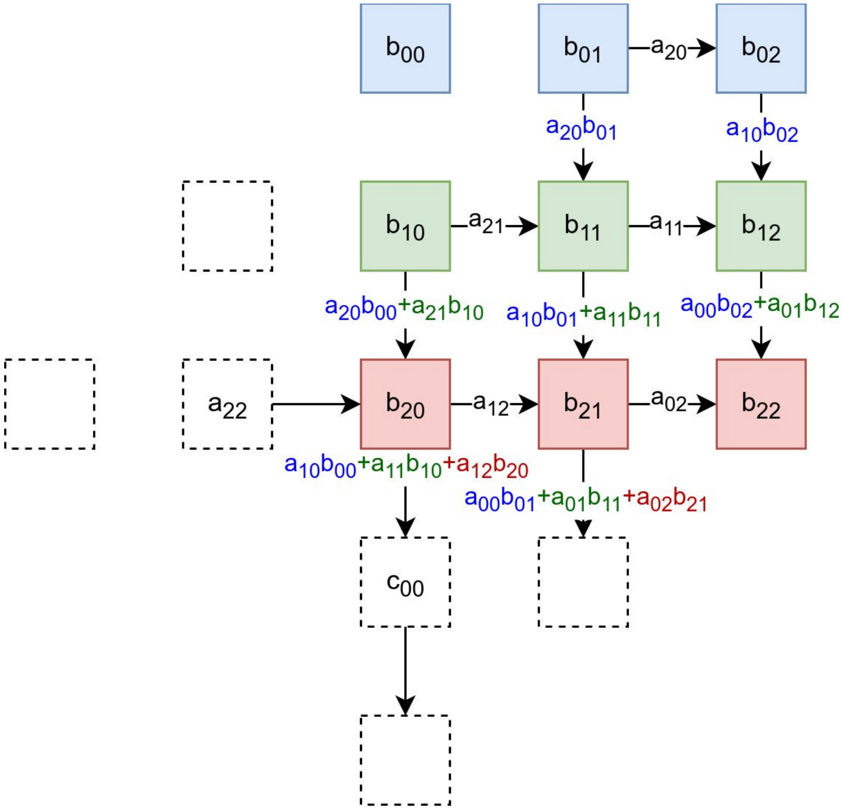

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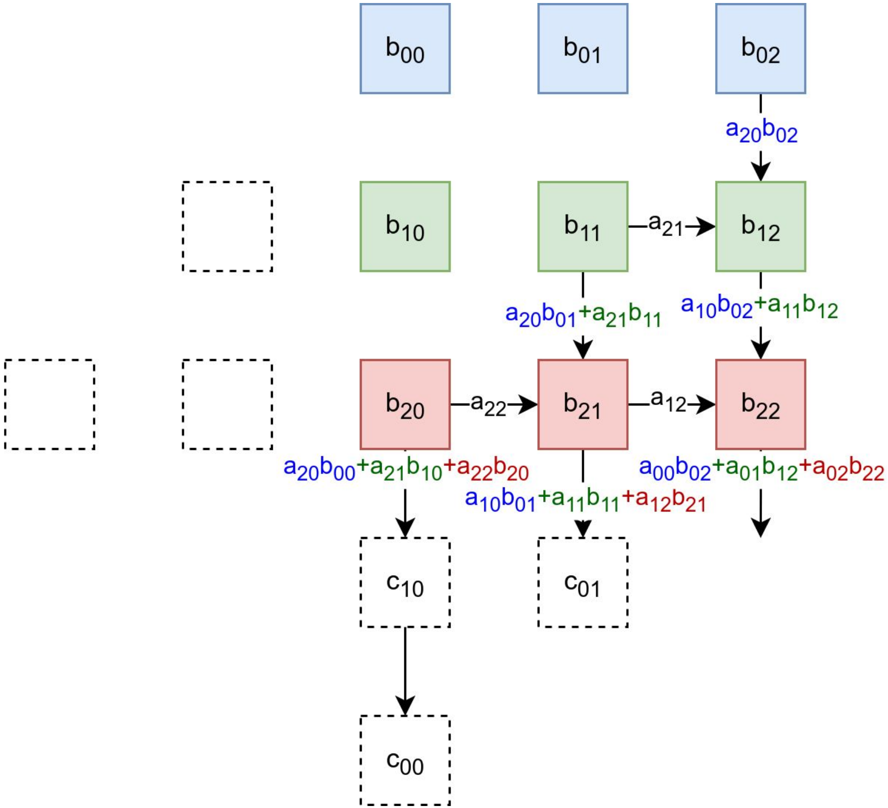
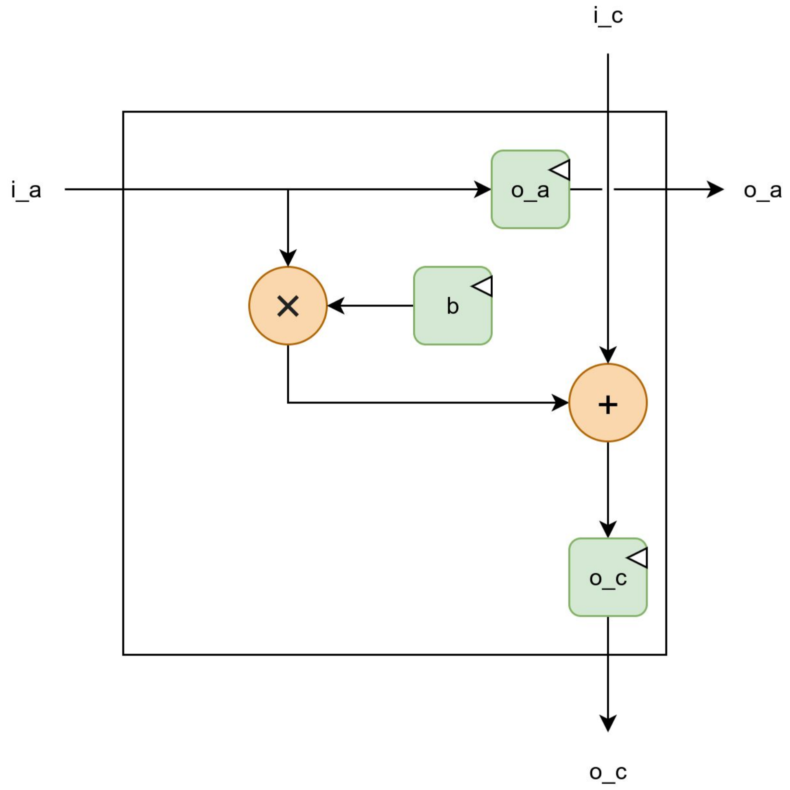
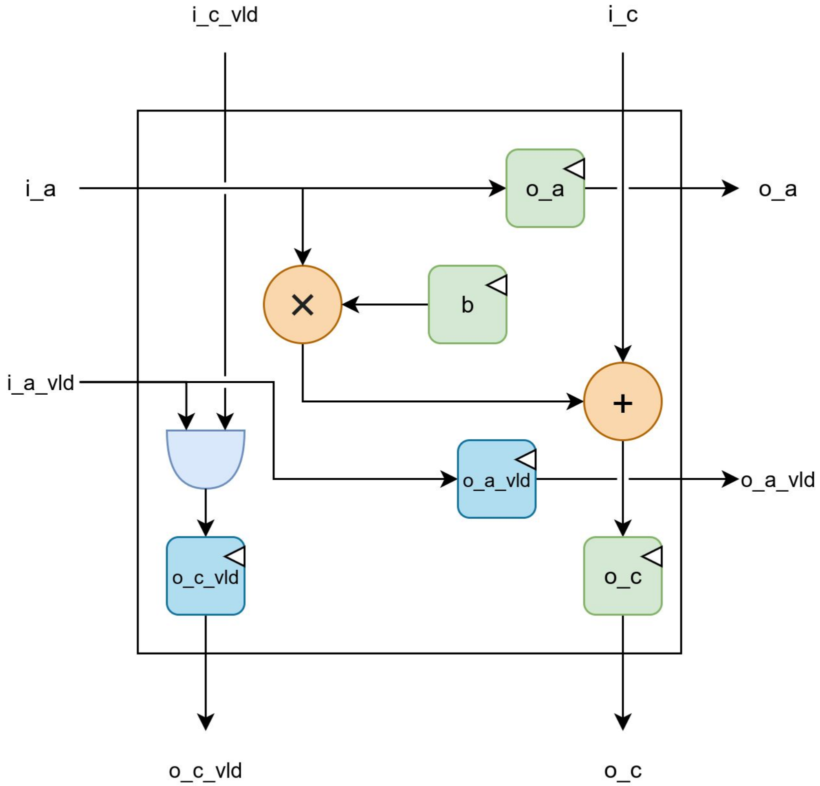


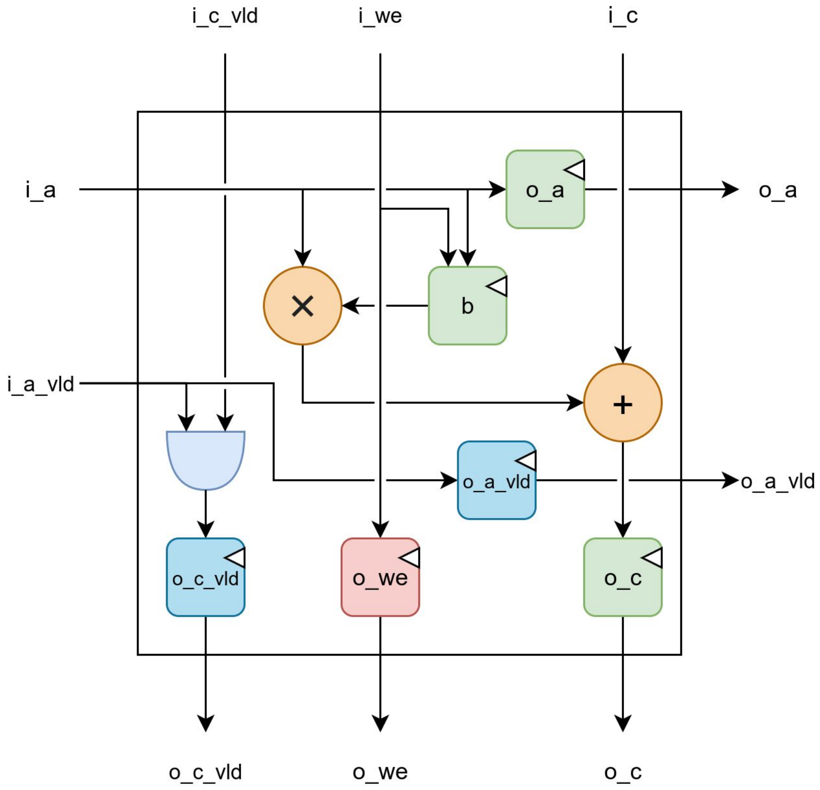
Processing Element: Data Flow



Processing Element: Control Flow



Processing Element: Write Enable



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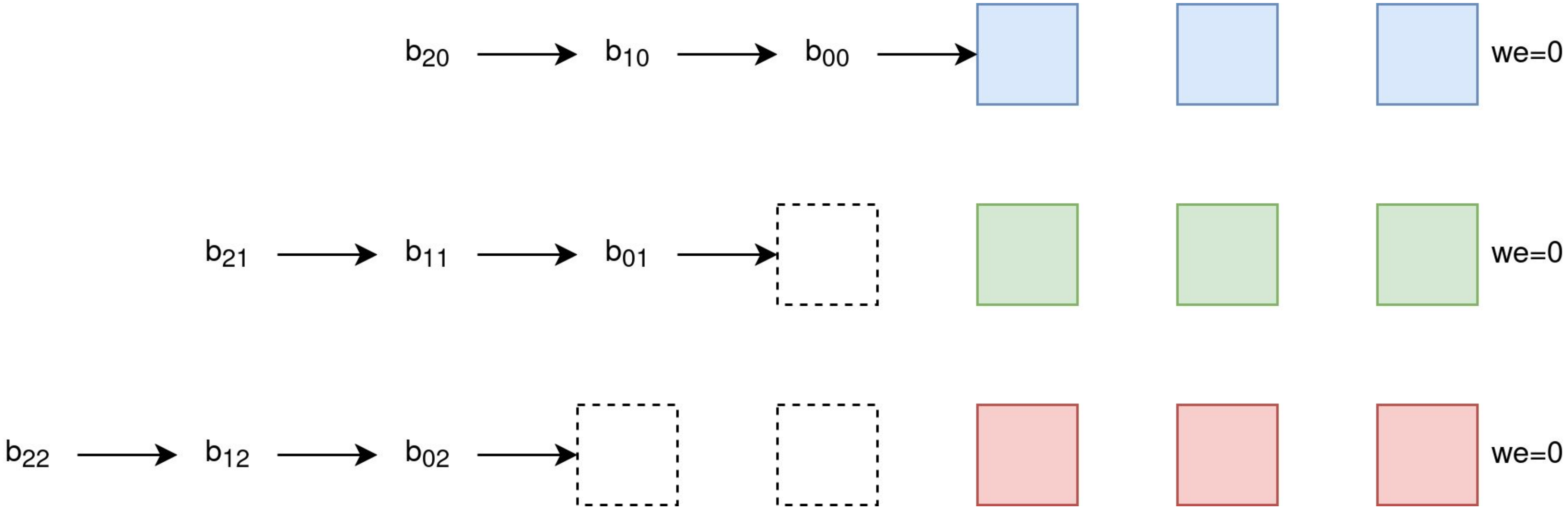
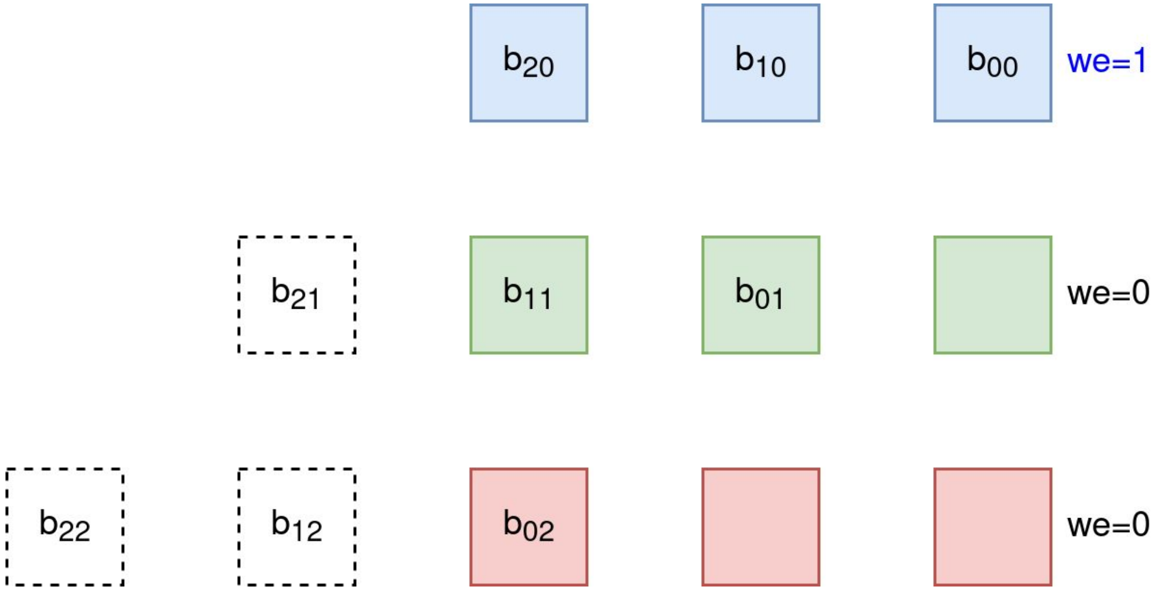
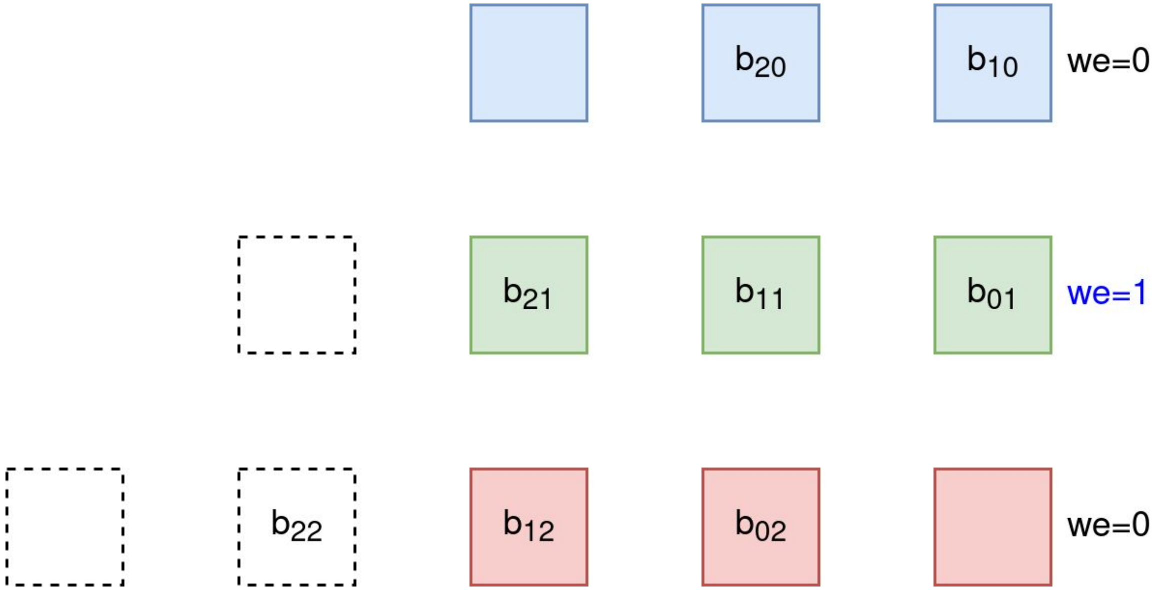


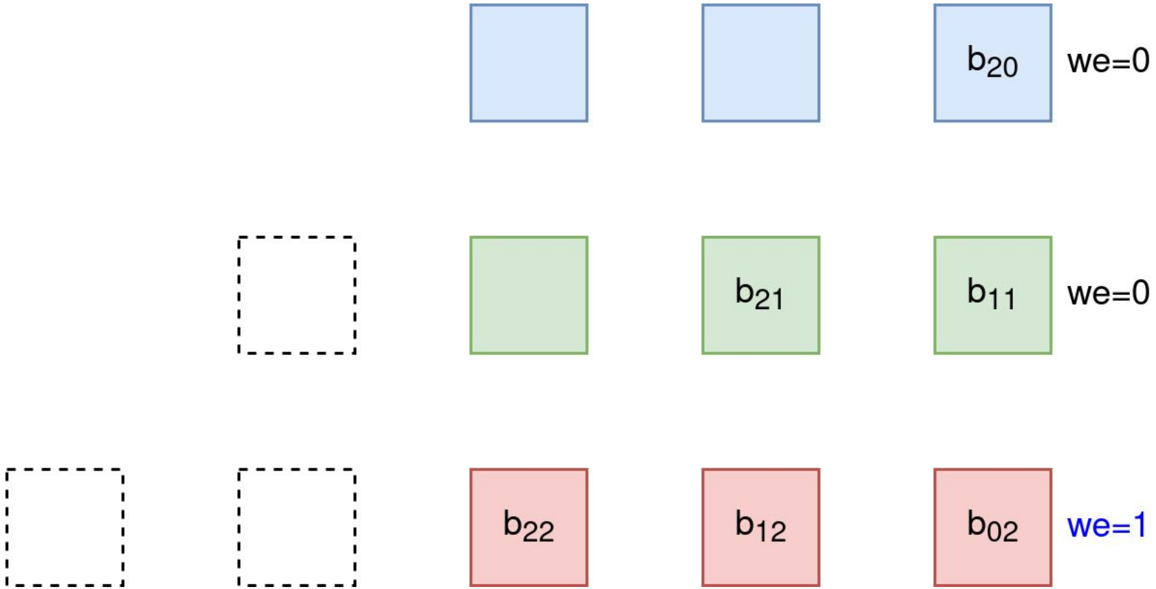
Systolic Array: Write Enable: 3



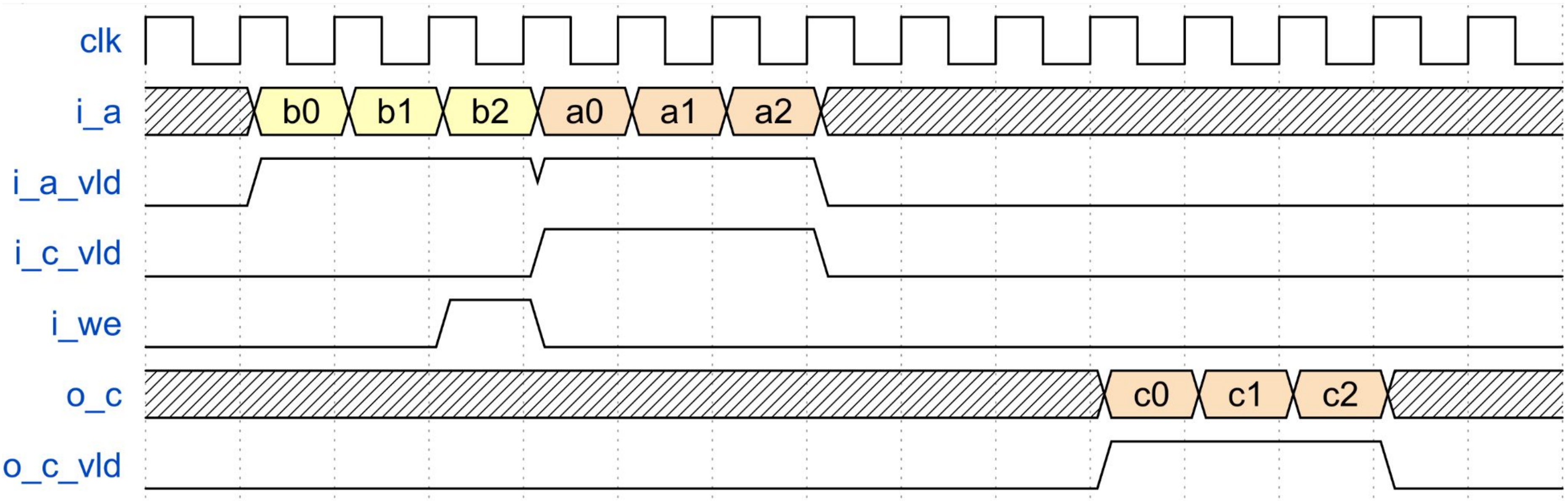
Systolic Array: Write Enable: 4



Systolic Array: Write Enable: 5



Systolic Array



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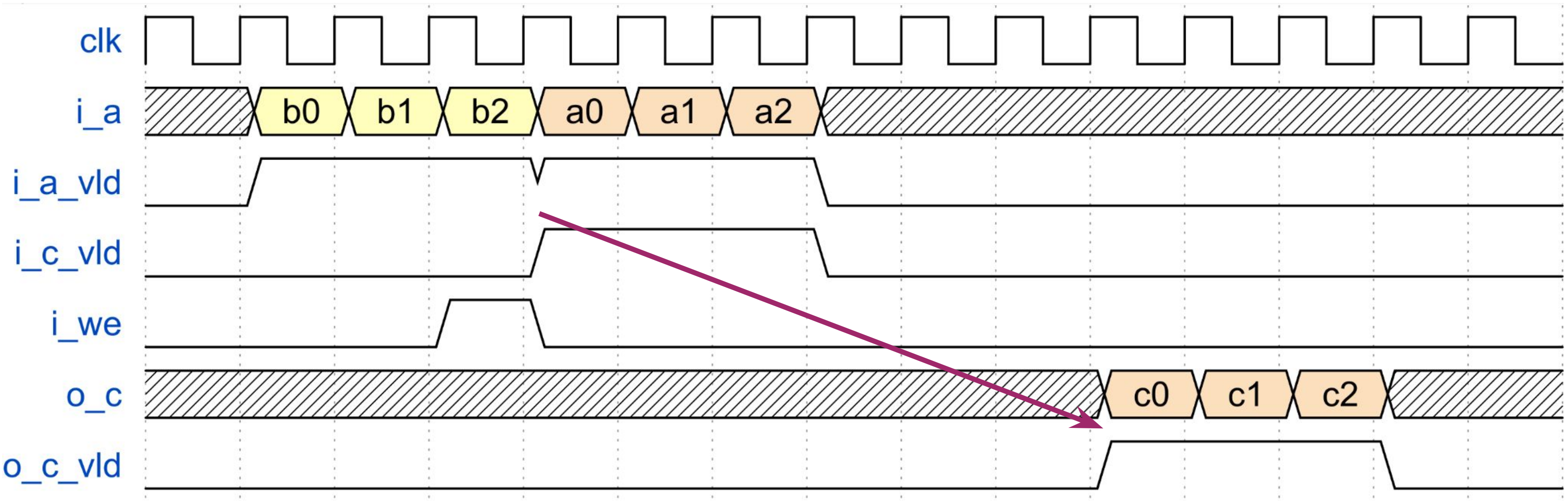
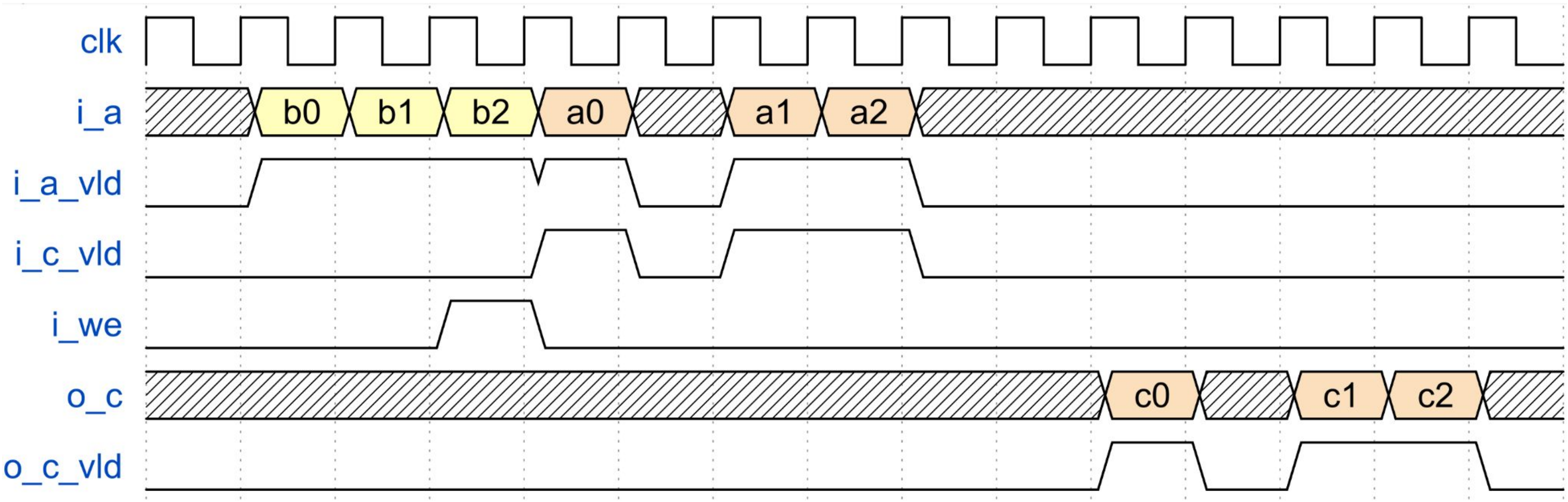
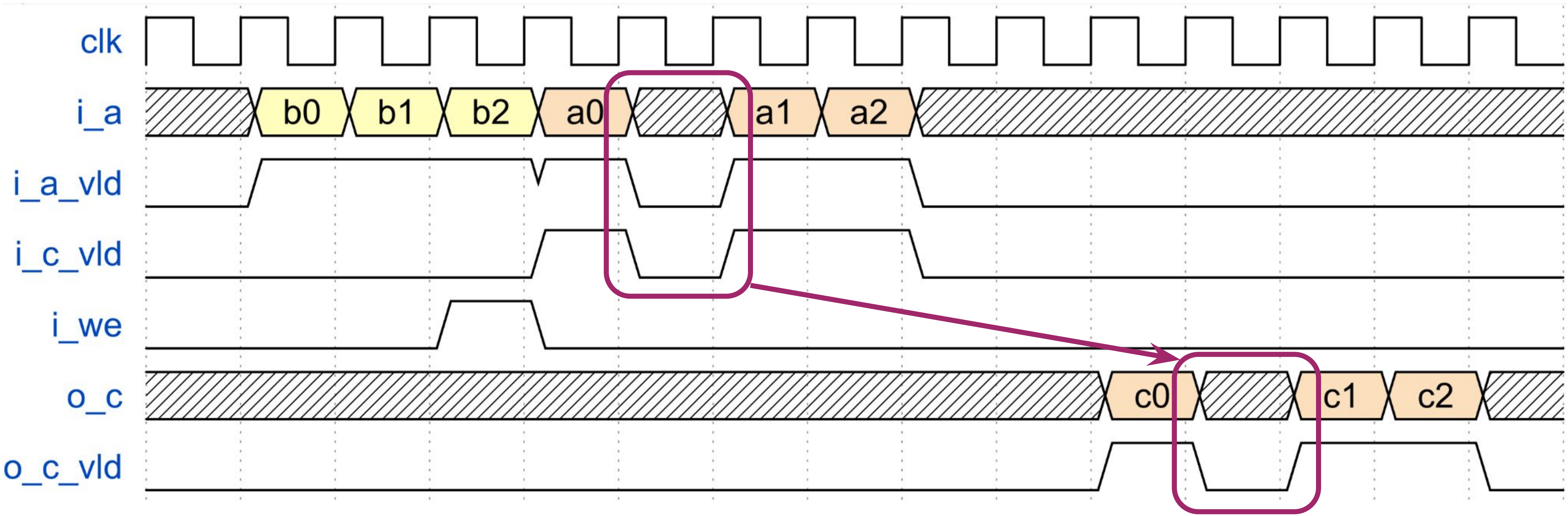


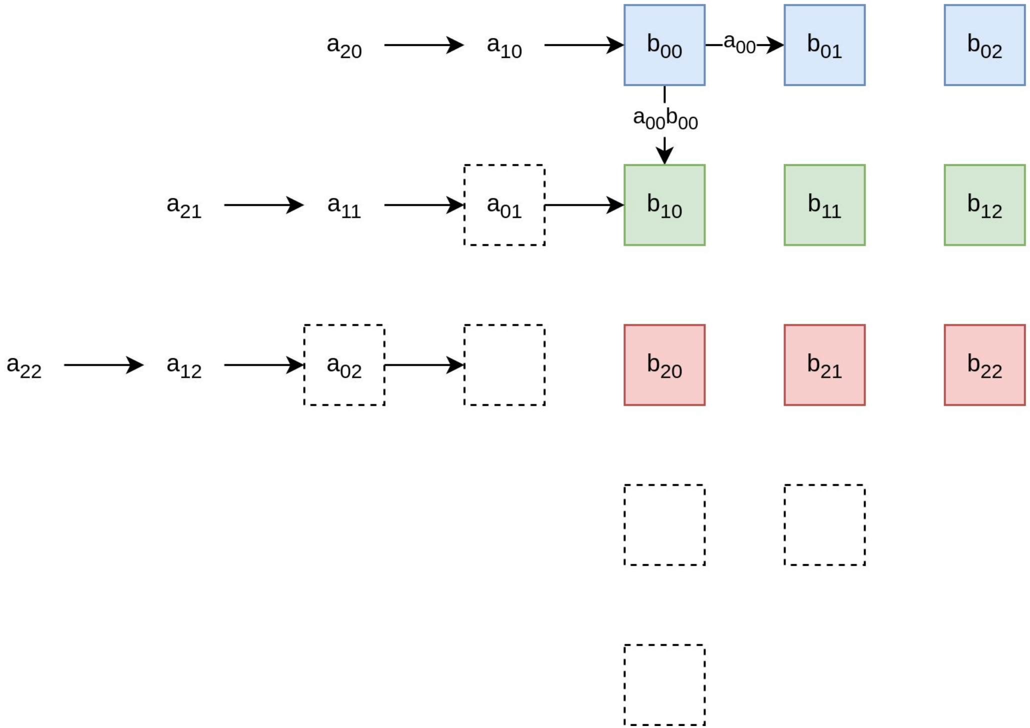
Systolic Array: Idle (A)



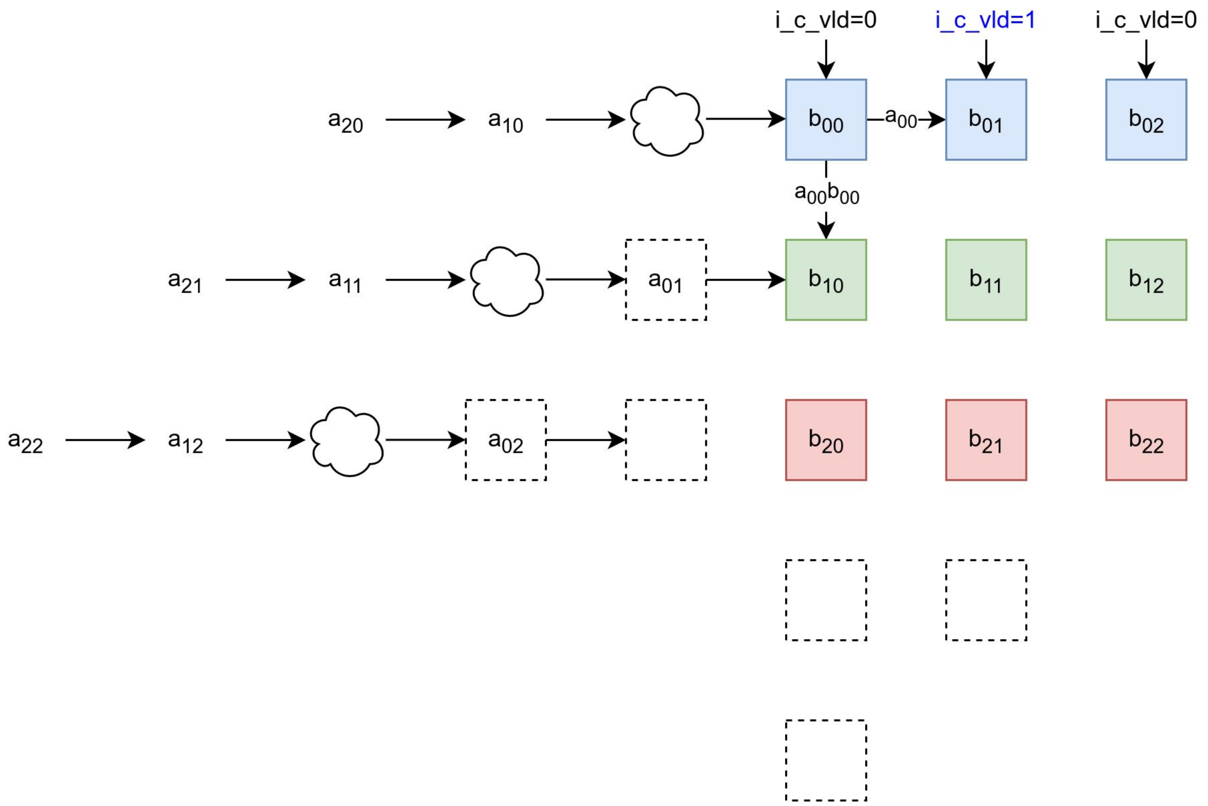
Systolic Array: Idle (A)



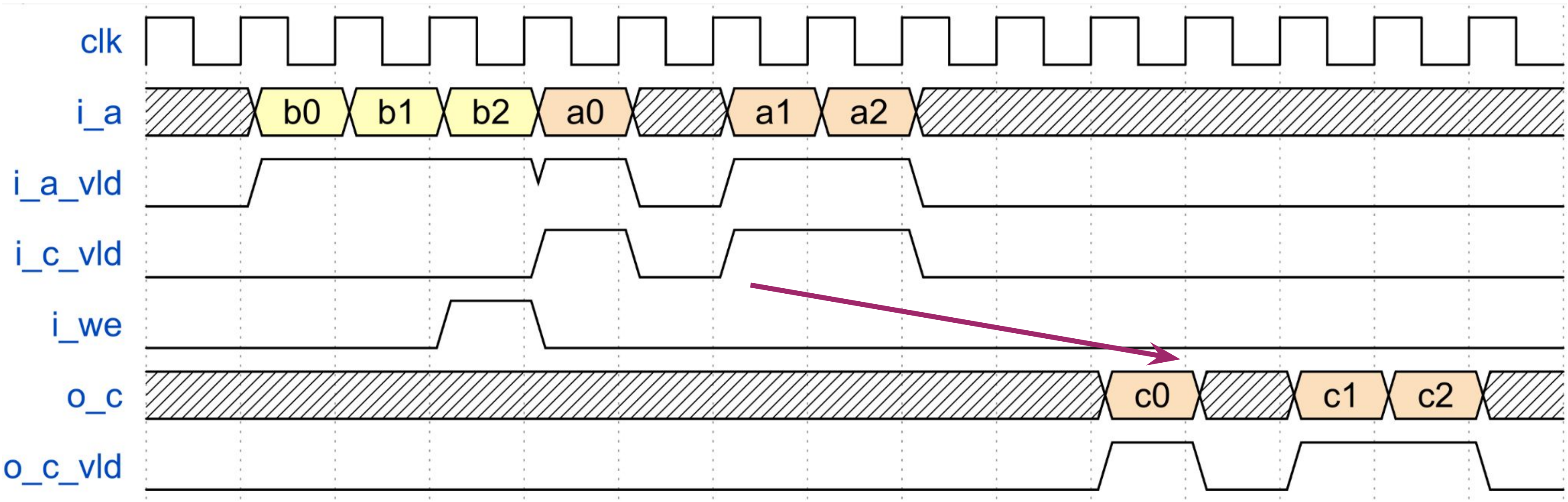
Systolic Array: Idle (A)



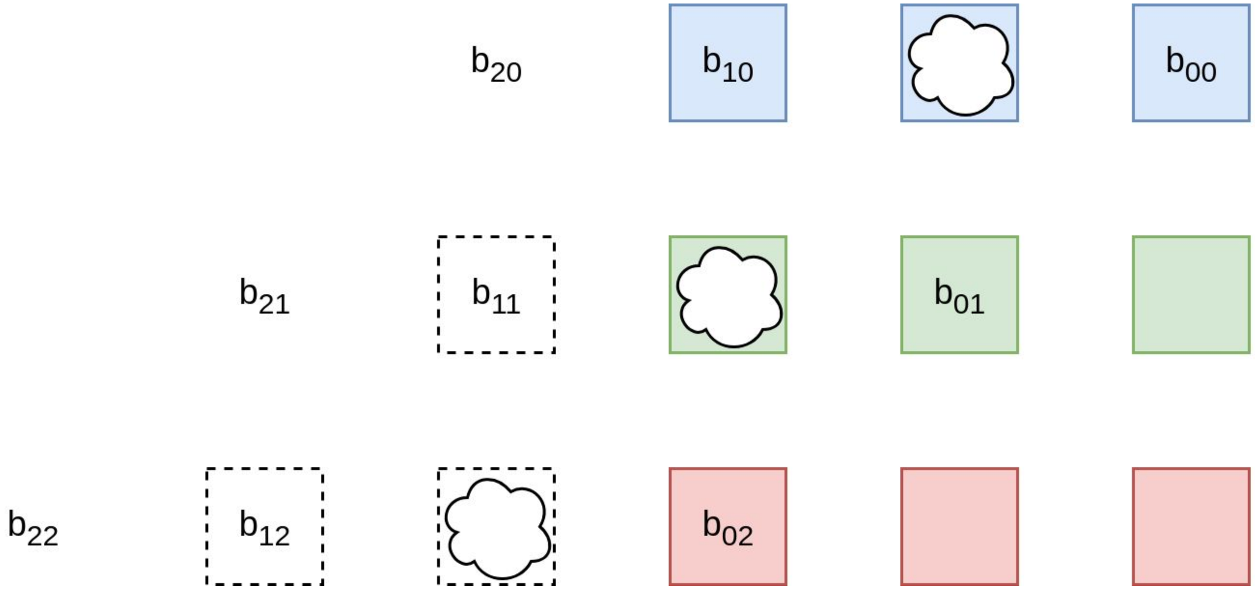
Systolic Array: Idle (A)



Systolic Array: Idle (A)



Systolic Array: Idle (B)



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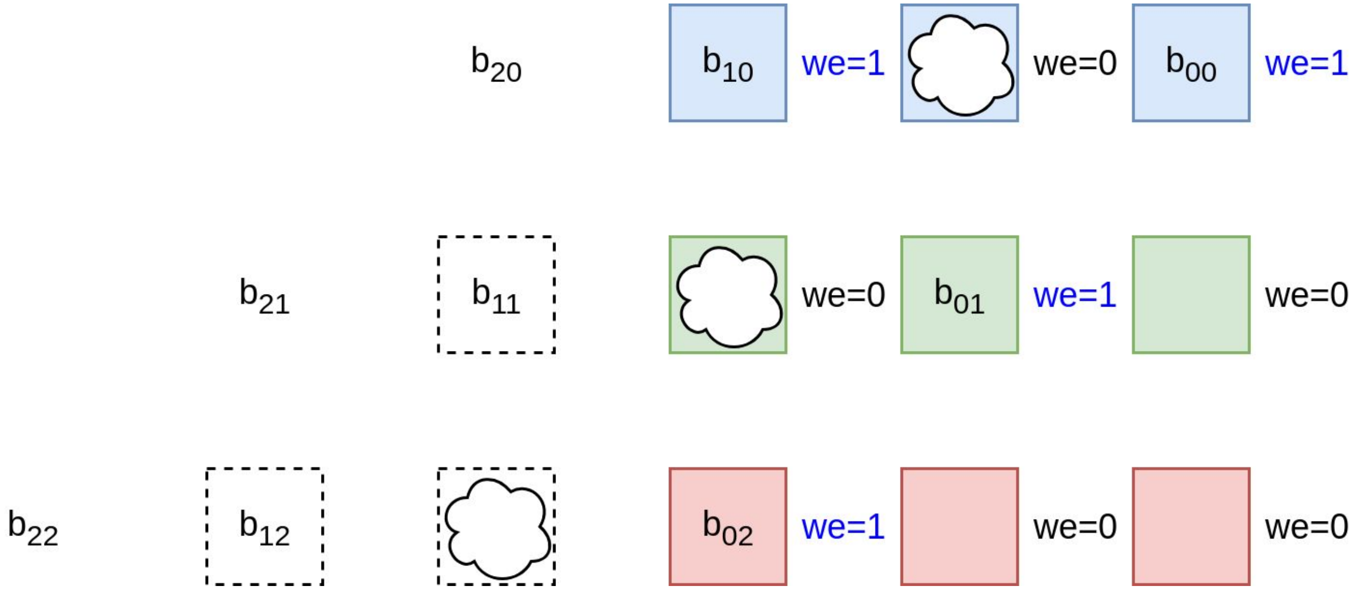
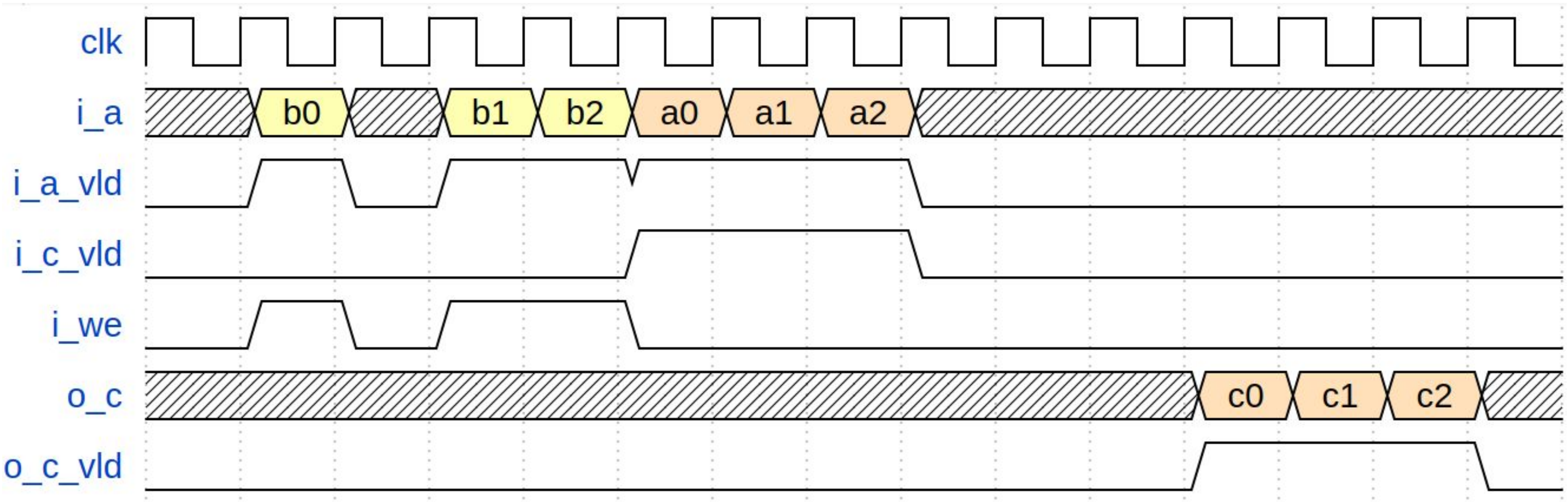


Systolic Array: Idle (B)



Задание

1. Processing Element, параметризованный разрядностью операндов (WIDTH=16)
2. Систолический массив, параметризованный размером (SIZE=4) и разрядностью
3. Сдвиговый регистр с valid-интерфейсами, параметризованный разрядностью
4. Систолический массив (WIDTH=16, SIZE=4) со сдвиговыми регистрами
5. Тестбенч для систолического массива, проверяющий умножение матриц m,n,k=4,4,4
6. Систолический массив с поддержкой Idle по A и B матрицам

Литература

- Jouppi et al., "In-Datacenter Performance Analysis of a Tensor Processing Unit", ISCA 2017
- H.T. Kung, "Why Systolic Architectures?," IEEE Computer 1982